

Biswajit Jana

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Research interests: exoplanet detection and characterisation, high-precision spectroscopy and radial velocity, astronomical instrumentation and closed-loop control, time-series analysis, interferometry, and reproducible astronomical data pipelines.

EDUCATION

University of Hertfordshire

Hatfield, United Kingdom

MSc Astrophysics with Advanced Research, Commendation

2022–2024

- **Dissertation:** *Closed-Loop Feedback Control System for the Exoplanet High-Resolution RV Spectrograph (EXOHSPEC) Development.*
- Advanced Research focus: high-resolution spectroscopy, exoplanet detection, astronomical instrumentation, statistical analysis and observational astronomy; practical CCD photometry/spectroscopy at Bayfordbury Observatory.

University of Engineering and Management

Jaipur, India

BTech Electronics and Communication Engineering, First Class with Distinction

2017–2021

- Dissertation: *Dimming Controlled Visible Light Communication System using Raspberry Pi.*
- Thesis work **published** in *Optical and Quantum Electronics*, Springer.
- CGPA: 8.62/10 (78.7% per institutional conversion).

PUBLICATIONS

Manuscripts submitted

- G. Cheng, D. J. Pinfield, B. Gauza, Z. Guo, R. Kurtev, J. Osses, H. R. A. Jones, B. Burningham, J. Borissova, E. Kovachky, A. Hamid, S. Harsora, C. F. P. Afonso, L. J. Hall, A. Scannell, O. Gifford, M. Belgrove, A. Gaffoor, B. Laurent, and **B. Jana**, “10 Benchmark T-Dwarfs: Wide Companions to Gaia Primaries,” submitted to *MNRAS* (2026).

Peer-reviewed publications

- S. Das, **B. Jana**, and S. K. Mandal, “Implementation of dimming controlled visible light communication using Raspberry Pi,” *Optical and Quantum Electronics* **53**, 725 (2021). doi:[10.1007/s11082-021-03362-4](https://doi.org/10.1007/s11082-021-03362-4).
- **B. Jana**, P. Roy, S. Sengupta, S. Porey, and R. Sen, “GSM controlled location specific garbage collecting Smart-bin,” *2019 9th Annual IEMECON*, pp. 211–214 (2019). doi:[10.1109/IEMECONX.2019.8877007](https://doi.org/10.1109/IEMECONX.2019.8877007).
- R. Sen, S. Das, G. G. Mazumder, P. Yadav, B. Neogy, R. Pandey, S. Sharma, and **B. Jana**, “Priority encoder using reversible logic gates in QCA,” *2017 8th IEEE IEMCON*, pp. 319–323 (2017). doi:[10.1109/IEMCON.2017.8117242](https://doi.org/10.1109/IEMCON.2017.8117242).

RESEARCH AND PROJECT EXPERIENCE

Closed-Loop Feedback Control System for EXOHSPEC

2023–Present

MSc Advanced Research / Visiting Researcher project; supervisors: Profs. Hugh R. A. Jones and William E. Martin

- Developed an end-to-end closed-loop stabilisation and telemetry pipeline for EXOHSPEC, an approximately $R \sim 70,000$ high-resolution radial-velocity spectrograph testbed, integrating IDS3010 interferometry, Meerstetter TEC control, BME/PT104 environmental sensing, MaxIm DL centroid tracking, LK220 cooling telemetry, and active-optics tip-tilt correction through Python.
- Quantified environmental coupling in the optical path: approximately $0.4 \mu\text{m}$ OPL per 1 hPa pressure change and approximately $6 \mu\text{m}$ OPL per 1°C temperature change; achieved TEC stability of $\sigma_T = 2.47 \text{ mK}$ with 17.4 mK peak-to-peak variation during a 44.5 h run.
- Commissioned a replacement active-optics unit at approximately $0.0194\text{--}0.0198 \mu\text{m}$ per step ($\sim 30\text{--}35\times$ finer than the previous actuator); long-duration V17 validation reached 88.59 h across 4,761 settled feedback frames with $dX_{\text{RMS}} = 0.0594 \text{ px}$ and $dY_{\text{RMS}} = 0.1631 \text{ px}$, with all settled measurements inside $\pm 0.5 \text{ px}$.
- Extended the detector pipeline to multi-line ThAr phase-correlation tracking, reaching $dX_{\text{RMS}} = 0.0373 \text{ px}$ and $dY_{\text{RMS}} = 0.0990 \text{ px}$ over 3.98 h. Using the current-setup dispersion of $0.01394 \text{ \AA px}^{-1}$, the dX RMS corresponds to $5.20 \times 10^{-4} \text{ \AA}$ of detector-equivalent spectral motion, or roughly $22\text{--}31 \text{ m s}^{-1}$ across $7000\text{--}5000 \text{ \AA}$; this is an instrument-stability scale, not a calibrated stellar RV precision.
- V19 thermal diagnostics combined 301 science frames with 3,612 LK220 measurements and identified a common $\sim 33.4 \text{ min}$ periodic component linking HVAC/thermal behaviour, chiller response, camera temperature and residual detector motion.
- Currently designing a 3D-printed CMOS detector mount with a controlled air-flow channel to test whether local convective/thermal gradients around the camera contribute to residual image motion; the planned experiment will compare airflow configurations against camera, LK220/PT104 and detector-centroid telemetry.
- Skills developed: precision optical metrology, feedback control, active optics, thermal diagnostics, experimental calibration, Python instrument control, time-series analysis, telemetry design, CAD/3D-print prototyping, and reproducible validation.

Dimming-Controlled Visible Light Communication System using Raspberry Pi

2019–2021

BTech thesis, University of Engineering and Management, Jaipur

- Designed and implemented a low-cost Raspberry-Pi visible-light communication prototype using multi-header hybrid pulse-position modulation (MH-HPPM), testing three dimming levels ($\alpha = 0.25, 0.50$ and 0.75) with simultaneous illumination and data transmission.

- Demonstrated communication to approximately 3.0 m at dimming 0.25 and 3.5 m at dimming 0.50/0.75, with throughput also tested against incidence angle; developed skills in optical wireless communication, embedded prototyping, modulation/decoding and experimental characterisation. [\[PublishedPaper\]](#)

ADDITIONAL ASTRONOMY RESEARCH EXPERIENCE

Case Studies in Astrophysics

University of Hertfordshire

MSc research-based coursework

Sep–Dec 2022

- **Exoplanets:** analysed transit dips and radial-velocity variations with DACE, Python and `lightcurve`; **brown dwarfs:** compared atmospheric-model interpretations for VHS 1256 b ($T_{\text{eff}} \sim 1100$ K, $\log g \sim 5.5$) and proposed PICASO follow-up.
- **Gaia close encounters:** used perihelion-distance and uncertainty analysis to identify GJ 710 as a potential close-passage candidate.

Bayfordbury Observatory Practical Experience

University of Hertfordshire

Observational astronomy, calibration and public outreach

2022–2024

- Practical work in CCD calibration, photometry, image acquisition, guiding, spectroscopy and public telescope/planetarium sessions using observatory instrumentation.

INTERNSHIPS, SUMMER AND WINTER PROGRAMMES

LUMI SPACE / SEPnet Summer Placement

United Kingdom

Evaluating Microbolometer Camera Performance for Satellite Laser Ranging

5 Aug–4 Sep 2024

- Completed a four-week link-budget study comparing four tracking-detector approaches (microbolometer, CMOS, CCD and quad detector) and built Python/Colab analyses around seven loss/performance terms: beam divergence, atmospheric turbulence/transmission, background light, optical losses, detector sensitivity and pointing error.
- Investigated wavelength/site-dependent extinction and detector improvements (high-TCR $\text{VO}_x/\text{a-Si}$, IR optical antennas, thermal isolation, pixel-size and NUC trade-offs); presented the final work at the SEPnet Student Expo, London, 20 Nov 2024.

Summer Research Project

University of Hertfordshire

Identifying Halo T Dwarf Candidates using VISTA, DES, and WISE Survey Data

Jun–Jul 2023

- Worked under Professor David Pinfield on cross-survey candidate vetting using VISTA, DES and WISE photometry/astrometry, including colour-based screening, wide-companion assessment and follow-up prioritisation.
- My contribution included candidate assessment and input to the successful CNTAC Magellan/FIRE follow-up proposal; the wider collaboration subsequently confirmed 10 benchmark T-dwarf/Gaia wide-companion systems, including eight previously unreported systems, forming the basis of the manuscript submitted to *MNRAS* in 2026.

Carl Sagan Exoplanet Summer Workshop

Caltech / NASA Exoplanet Science Institute

Atmospheric modelling and exoplanet analysis training

2023

- Gained hands-on exposure to exoplanet-atmosphere modelling, brown-dwarf atmosphere interpretation, and retrieval tools including PICASO and petitRADTRANS.

Radio Astronomy Summer School

Gauribidanur Radio Observatory / Raman Research Institute / Naxxatra

Radio instrumentation and interferometry

Feb–May 2022

- Worked under Dr. Ramesh Balasubramanyam on a two-element interferometer, solar-transit observations and receiver-design concepts around 1.4 GHz with approximately 100 MHz bandwidth, covering baselines, phase/coherence and receiver chains.

IIT Kharagpur Winter Internship

India

Electronics, FPGA sensing and multi-drone synchronisation

22 Dec 2019–23 Jan 2020

- Four-week research internship under Professor Sudip Misra on low-cost high-frequency detection circuits using FPGA hardware and calibration/synchronisation concepts for multi-drone swarm configurations.

SELECTED SCIENTIFIC SOFTWARE AND OPEN-SOURCE PROJECTS

Finding Earth 2.0 – Reproducible Exoplanet Population and Evidence Pipeline

2026–Present

Public-archive exoplanet population analysis with explicit provenance and uncertainty propagation

- Ingested 164,209 provenance-tracked source records from 13 archive tables and consolidated 6,354 confirmed planets across 4,764 host systems; the strict screen retained 174 conservative-habitable-zone worlds and 15 small, temperate candidates.
- Propagated published uncertainties with 4,000 Monte Carlo draws per planet, retained 89,131 measurement-level provenance links across 1,814 publications, and separated measured masses from radius-inferred masses rather than mixing them.
- Skills developed: exoplanet population analysis, archive-scale Python pipelines, catalogue cross-matching, uncertainty propagation, provenance design, scientific validation and reproducible reporting. [\[GitHub\]](#)

EXOhSPEC ThAr Atlas – Detector-Space Calibration Study

2026–Present

Measured thorium–argon FITS analysis from the EXOhSPEC laboratory testbed

- Analysed four measured ThAr exposures (30, 60, 120 and 180 s), detected 24 strong spatially separated features in the 120 s frame, and measured common unsaturated response linearity of $R^2 = 0.999276$ with a maximum fractional residual of 1.994%.
- Ran 1,000 seeded centroid injection–recovery trials per feature: pixelwise HDR centroiding reduced all-feature RMSE from 0.008338 px to 0.004675 px (43.9%) and strong-feature RMSE from 0.012954 px to 0.002086 px (83.9%).
- Skills developed: FITS detector analysis, calibration-lamp spectroscopy, centroid estimation, exposure optimisation, injection–recovery validation and explicit separation of detector coordinates from wavelength/RV claims. [\[GitHub\]](#)

ExoLight Transit Lab / ExoIntel-Prime

2026–Present

Exoplanet transit-photometry, forward-modelling and uncertainty-diagnostics pipeline

- Built a reproducible transit-analysis framework combining archival/local light curves with physical forward models using quadratic limb darkening, numerical stellar-disk integration, finite exposure-time integration and residual diagnostics.
- Executed 4,000 seeded injection–recovery trials across white and AR(1) correlated noise: for a 1,000 ppm transit, naive 95% coverage fell from 0.943 to 0.549 under correlation while event-cluster coverage remained 0.932; for a 300 ppm correlated-noise transit, naive three-sigma recovery was 0.980 versus 0.612 with event-cluster uncertainty.
- Applied the workflow across a portfolio of hot-Jupiter, Neptune and super-Earth systems spanning transit, phase-curve and atmospheric-interpretation case studies; maintained explicit separation between measured data, forward models and hypothesis demonstrations. [\[Project and report collection\]](#)
- Skills developed: transit physics, astronomical time-series analysis, uncertainty diagnostics, model–data comparison, reproducible scientific software and validation.

DESI/2MRS Survey-Scale Cosmology Pipeline (NĀSADĪYA LIGHTCONE)

2026–Present

Provenance-first processing of measured galaxy-survey data

- Processed 6,093,818 accepted DESI DR1 LSS rows across 4,205 spatial tiles plus the complete 43,533-row 2MRS local layer, producing a validated 185.12 MiB Parquet research product with source checksums, manifests and survey-integrity audits.
- Built a deterministic 125,000-row level-of-detail layer for interactive exploration while preserving the full research-scale data separately; skills developed include FITS/Parquet conversion, cosmological coordinate transforms, multi-million-row pipeline design, provenance control and reproducible figure generation. [\[GitHub\]](#)

Gaia Galactic Explorer & H–R Engine

2026–Present

Gaia DR3 catalogue analysis and stellar-population visualisation

- Queried Gaia DR3 through TAP/ADQL and constructed colour–magnitude and local 3-D stellar-distribution products from measured parallax and photometry, computing distance and absolute Gaia *G* magnitude directly from catalogue fields with no synthetic fallback stars.
- Redesigned the ingestion workflow after a 50,000-row single-download failure by partitioning larger queries into RA slices; skills developed include catalogue filtering, stellar-population analysis, chunked data ingestion, scientific visualisation and performance debugging. [\[GitHub\]](#)

TALKS AND POSTER PRESENTATIONS

- *Testing Performance of Microbolometer Detector for Satellite Laser Ranging Application*. SEPnet Student Expo, London, 20 Nov 2024.
- *High Resolution RV Spectrographs: ANDES and PID Loop Implementation in EXOhSPEC*. University of Hertfordshire, 7 Mar 2024. [\[Slides\]](#)
- *Optimising Path Length Stability in Laser Interferometers using Air Refractive Index*. University of Hertfordshire, 8 Feb 2024. [\[Slides\]](#)
- *Advancement in Precision: Laser Interferometer Control System*. University of Hertfordshire, 8 Dec 2023. [\[Slides\]](#)
- *State of the Art: Radial Velocity Spectrograph*. University of Hertfordshire, 9 Nov 2023. [\[Slides\]](#)

TECHNICAL SKILLS

- **Instrumentation and optics:** laser optics, interferometry, IDS3010 displacement sensing, radial-velocity spectrograph stabilisation, environmental control, optical alignment, active-optics tip–tilt correction, TEC/PID control, thermal diagnostics, CAD and 3D-printed experimental hardware.
- **Scientific computing and data handling:** Python for instrument control, telemetry and data-processing pipelines; NumPy, SciPy, pandas, Matplotlib, Astropy, lightcurve, DACE, FITS, Parquet, CSV/JSON, archive/API workflows, Jupyter, time-series analysis, uncertainty propagation, data validation and provenance tracking.
- **Astronomical data and modelling:** exoplanet transit and radial-velocity analysis, high-resolution spectroscopy, ThAr calibration-lamp analysis, Gaia DR3/TAP/ADQL, DESI/2MRS survey products, catalogue cross-matching, Monte Carlo propagation, residual diagnostics and injection–recovery testing.
- **Telemetry and control systems:** Grafana, InfluxDB, serial communications, RS-232/RS-485, sensor fusion, real-time logging, feedback-control dashboards, multi-sensor thermal/environment monitoring and closed-loop diagnostics.
- **Hardware and embedded systems:** Arduino, Raspberry Pi, NodeMCU ESP8266, PSoC 5LP, embedded boards, IoT systems, visible-light communication prototypes, detector/sensor integration and laboratory electronics.
- **Scientific software and pipeline development:** JavaScript, HTML/CSS, WebGL/Canvas, browser workers, chunked catalogue loading, interactive scientific visualisation, GitHub Pages, reproducible research interfaces and large-data pipeline debugging.
- **Additional programming, software and communication:** MATLAB/Simulink, C/C++, PHP, SQL, Bash, MaxIm DL, Aperture Photometry Tool, RSpec, QCA Designer, PSpice, PSoC Creator 5LP, Git/GitHub and LaTeX/Overleaf; technical reports, posters, telescope proposals, public research visualisation and methods/provenance documentation.